Verrucous Epidermal Nevus

Verrucous epidermal nevus is also known as linear verrucous epidermal nevus or linear epidermal nevus.

Clinical Features

Verrucous epidermal nevi are characterized by localized or diffuse, closely set, skin-colored, brown, or gray-brown verrucous papules, which may coalesce to form well-demarcated papillomatous plaques. Linear configurations are common on the limbs, often following Blaschko's lines or relaxed skin tension lines.

Extensive distribution of a verrucous epidermal nevus is termed **systemized epidermal nevus**. Variants include: - **Nevus unius lateris**: epidermal nevi affecting one-half of the body - **Ichthyosis hystrix**: bilateral, widespread epidermal nevi

Systematized nevi typically have a transverse orientation on the trunk and a linear pattern on the limbs.

An epidermal nevus presenting with pruritus, erythema, and scaling is likely an **inflammatory linear verrucous epidermal nevus (ILVEN)**. These lesions most commonly occur on the buttocks and lower extremities and may resemble linear psoriasis.

Course and Complications

Linear epidermal nevi usually appear between birth and adolescence. Congenital lesions tend not to enlarge significantly, whereas postnatally appearing lesions may grow during childhood and stabilize around puberty.

Most epidermal nevi remain quiescent after adolescence. However: - Intertriginous lesions may become macerated and secondarily infected - Rare cases of **basal cell carcinoma** and **squamous cell carcinoma** arising within epidermal nevi have been reported - Malignant transformation occurs most often in middle-aged or elderly individuals, though rare cases (e.g., a 17-year-old woman) have been documented

Epidermal nevi may coexist with other cutaneous findings such as: - Café-au-lait macules - Congenital hypopigmented macules - Congenital nevocellular nevi

Extensive lesions may be associated with extracutaneous abnormalities, part of the **epidermal nevus syndrome**.

Rarely, patients with epidermal nevi have offspring with **epidermolytic hyperkeratosis (EHK)**, caused by mutations in keratin 10 (K10). Studies show that affected parents exhibit mosaicism due to postzygotic mutations—mutations are present in the nevus tissue but not in normal skin. Their children inherit the same K10 mutation in all cells.

If histopathology of an epidermal nevus shows features of EHK, the patient has an increased risk of passing EHK to offspring.